

Machine Health Intelligence (MHI)

AI-Powered Predictive Maintenance Report

Machine Parameters

Machine Type: L
Air Temperature: 300.0 K
Process Temp: 310.0 K
Rotational Speed: 1500 RPM
Torque: 100.0 Nm
Tool Wear: 299 min

Prediction Summary

Failure Probability : 90.89%
Health Score : 9.11%
Risk Level : Critical
Estimated Downtime : 6–8 Hours
Estimated Cost : ■30,000+

Failure Causes

- High Torque
- Excessive Tool Wear

Maintenance Recommendations

1. Stop machine immediately.
2. Perform complete mechanical inspection.
3. Replace worn cutting tool.
4. Check and service the cooling system.
5. Inspect spindle bearings.
6. Reduce operating load before restart.

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